

SA-2 EVENT

LINUX RECOVERY • SECURITY • CONTAINER CHALLENGE

RECOVER | SECURE | DEPLOY

INTRODUCTION

REAL-WORLD SCENARIO

You have been assigned as a Linux System Administrator for a company.

The Head of the company has contacted you regarding a critical problem with one of their Linux servers. One of the company staff members made several changes to the operating system, and as a result, the system is no longer functioning correctly.

The server has multiple configuration issues, and in its current state, you are unable to access the operating system normally.

Your responsibility is to take control of the situation and bring the server back to a fully working condition.

The company management has provided you with a list of requirements that the server must meet after the recovery and troubleshooting process.

STEP 1 – DOWNLOAD AND START THE OPERATING SYSTEM

The first task is given below.

Download the required Linux Operating System from the Google Drive link provided below.

At present, the operating system is not accessible normally, and you will not be able to log in to the system using the regular login process.

Login Information

- Username: root
- Password: 1234

Note: The above credentials are provided for this practical lab environment only.

Task

1. Download the Operating System from the Google Drive link.
2. Set up the operating system in your lab environment.
3. Attempt to access the system using the credentials provided above.
4. Identify the issue preventing normal access to the operating system.
5. Proceed with the troubleshooting steps provided in the following sections.

Google Drive Link:
https://drive.google.com/file/d/1Cgvz8nxgdLTe0GrOpR5Qm6PfOFj0AiPo/view



STEP 2 – SYSTEM SECURITY, WEB SERVER & PERFORMANCE CONFIGURATION

After gaining access to the operating system, the company has provided the following requirements.

You are responsible for implementing all the configurations below and verifying that they work correctly.

Task 2.1 – Configure SSH Security

For security purposes, the company wants to change the SSH service port from the default port to a new port.

Requirements

- Change the SSH service from port 22 to the new port specified by the company.
- Ensure that the SSH service is configured correctly on the new port.
- Configure the system so that a new user can connect to the server only through the new SSH port.
- Make sure the configuration works correctly with SELinux enabled.
- Verify the configuration after restarting the SSH service.
- Ensure that the configuration remains persistent after reboot.

Task 2.2 – Configure the Apache Web Server

The company wants to restrict the content that can be accessed through the Apache web server.

Create a dedicated directory named:

/companyweb

Requirements

- Create the /companyweb directory.
- Only the files placed inside /companyweb should be accessible through the Apache web server.
- Configure Apache to use /companyweb as the required web root.
- Configure the appropriate SELinux file context for the directory.
- Ensure that Apache is able to access and serve the content correctly.
- Verify that the web page can be accessed through a browser or using an appropriate command-line test.
- Ensure that the configuration remains persistent after reboot.

Task 2.3 – Configure System Performance Profile

The company wants the server to operate using a specific power-management profile.

Requirements

- Configure the system to use the Power Save tuning profile.
- Verify that the required tuning profile is active.
- Ensure that the configuration is persistent.

Task 2.4 – Monitor System Performance

The company wants to monitor the server's CPU and memory usage.

When the system experiences high CPU or memory utilization, the administrator should be able to identify:

- Which process is consuming the resources.
- The CPU and/or memory usage.
- The time at which the issue occurred.
- Relevant information about the process.

The collected information must be stored in a log file for later analysis.

Task 2.5 – Create a Monitoring Script

Create a Shell Script that monitors system performance.

The script should:

1. Monitor CPU and memory usage.
2. Detect when resource usage exceeds the defined threshold.
3. Identify the process responsible for the high resource usage.
4. Record the relevant information in a log file.
5. Include the date and time of the event in the log.
6. Continue monitoring according to the requirements of the company.

Log File

Store the monitoring information in an appropriate log file, for example:

/var/log/system-monitor.log

Task 2.6 – Schedule the Monitoring Script

The server does not need to be monitored throughout the entire day.

According to the company:

The server is required only between 6:00 PM and 8:00 PM every day.

Therefore, the monitoring script must be scheduled so that it runs only during this period.

Requirements

- The monitoring script must automatically start at 6:00 PM.
- It must monitor the system during the required period.
- It must automatically stop at 8:00 PM.
- The monitoring should repeat automatically every day.
- The solution must be persistent and continue working after a system reboot.

STEP 3 – STORAGE, SWAP & EXPANDABLE PARTITION

The company has added two additional hard disks to the Linux server.

The server currently has 1.5 GB of RAM. The company wants to increase the available swap space and also requires additional storage for important data.

Your task is to configure the additional storage according to the requirements below.

Task 3.1 – Increase Swap Space

The server currently has 1.5 GB of RAM.

The company requires an additional 1 GB of swap space.

Requirements

- Create an additional 1 GB swap space using the available storage.
- Enable the swap space.
- Verify that the new swap space is active.
- Configure the system so that the swap space remains available after reboot.
- Verify the final swap configuration.

Task 3.2 – Create an Expandable Data Storage

The company needs a separate 1 GB storage area for important data.

Requirements

- Create a 1 GB data storage area using the available disk.
- Create an appropriate file system on it.
- Mount it at a suitable location.
- Add some important company data to this storage.
- Make the mount configuration persistent across reboot.

Task 3.3 – Prepare the Storage for Future Expansion

The company expects the amount of data stored in this location to increase in the future.

Therefore, the 1 GB storage must be designed so that it can be extended later without losing the existing data.

Requirements

- The initial storage size must be 1 GB.
- Add sample data to the storage before performing any extension.
- Extend the storage to a larger size when additional space is required.
- The existing data must remain intact after the extension.
- Resize the file system appropriately after extending the storage.
- Verify that the additional space is available and the existing data is still accessible.

STEP 4 – CREATE AN ASTRA LINUX CONTAINER IMAGE FROM THE EXISTING SYSTEM

The company has given you one of the most important tasks in this practical challenge.

You are currently working on a customized Linux operating system called ASTRA Linux, which is based on Rocky Linux.

The company now wants you to take the existing ASTRA Linux userspace environment and convert it into a container image.

The objective is to preserve the required files, directories, installed packages and configurations of the existing ASTRA Linux environment and make them available inside a container.

The resulting image must then be used to create and run a container on the same ASTRA Linux server.

Important: A container is not a virtual machine. The container will use the host system's Linux kernel. The objective of this task is to reproduce the required ASTRA Linux userspace environment inside a container.

Task 4.1 – Prepare the ASTRA Linux Filesystem

Prepare the required filesystem of the existing ASTRA Linux system to be used as the root filesystem of a container image.

Requirements

The container image should contain the required:

- System directories

- Configuration files
- Installed packages
- User and group information
- Custom directories
- Required application files
- Other required ASTRA Linux userspace components

You must ensure that the required ASTRA Linux environment is available inside the resulting container.

Task 4.2 – Create the Container Image

Convert the prepared ASTRA Linux filesystem into a container image.

Create the image with the following name:

astra-linux:latest

Requirements

- Create a suitable root filesystem archive.
- Import the filesystem into Podman as a container image.
- Assign the image name astra-linux:latest.
- Verify that the image has been successfully created.
- Inspect the image and confirm that it contains the required ASTRA Linux userspace environment.

Podman's import operation can create a filesystem image from a root-filesystem tarball, which is the mechanism relevant to this task.

Task 4.3 – Create a Container from the ASTRA Image

Use the newly created astra-linux:latest image to create a container.

The container must run on the same ASTRA Linux machine.

Requirements

- Create a container from astra-linux:latest.
- Give the container an appropriate name, for example:

astra-container

- Start the container successfully.

- Verify that the container is running.
- Verify that the required ASTRA Linux files, directories and configurations are available inside the container.

Task 4.4 – Configure SSH Inside the Container

The company wants administrators to be able to access the ASTRA Linux container through SSH.

Configure an SSH service inside the container.

Requirements

- Ensure that the required SSH packages are available inside the container.
- Configure SSH correctly.
- Create or configure a suitable user for SSH access.
- Ensure that the SSH service is listening on the appropriate container port.
- Verify that SSH is working correctly from inside the container.

Task 4.5 – Configure Port Mapping

The company has specified that the container must be accessible through port 4444 on the ASTRA Linux host.

The SSH service inside the container should use port 22.

Configure the following port mapping:

Host Port 4444 → Container Port 22

For example, the concept should be:

ASTRA Linux Host:4444
↓
ASTRA Container:22
↓
SSH Service

Podman supports publishing a container port to a different host port using port mapping.

Task 4.6 – Access the ASTRA Container Through SSH

From the ASTRA Linux host machine, connect to the container using SSH.

Requirements

- Use the host's port 4444.
- Establish an SSH connection to the container.
- Verify that the connection reaches the container, not the host operating system.
- Check the environment from inside the container.
- Verify the required ASTRA Linux files and directories.
- Verify that the required packages and configurations are available.
- Exit from the container successfully.

Task 4.7 – Verify the ASTRA Linux Environment

After connecting through SSH, verify that the required ASTRA Linux environment has been successfully reproduced inside the container.

Check the following:

- Operating system information
- Hostname/container identity
- Installed packages
- Required users and groups
- Important configuration files
- Custom directories
- Application files
- Required environment settings

The purpose of this verification is to confirm that the container contains the required ASTRA Linux userspace environment.

FINAL CHALLENGE

You are expected to complete the entire process without being given the exact commands.

Your workflow should be:

ASTRA Linux

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Prepare Required Filesystem

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Create Root Filesystem Archive

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Import as Container Image

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Create astra-linux:latest Image

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Create astra-container

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Configure SSH

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Map Host Port 4444 → Container Port 22

↓

SSH into the Container

↓

Verify ASTRA Linux Environment

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OBJECTIVE

The final objective is to demonstrate that you can take the required ASTRA Linux userspace environment, create a container image from it, run that image as a container, configure SSH access, and access the resulting ASTRA Linux environment remotely through port 4444.

This is a practical containerization and troubleshooting task. Do not simply follow commands. Understand what you are doing at every stage.

FINAL SUBMISSION REQUIREMENT

After completing all the assigned tasks, prepare a short and professional report for the Company Director.

1. **Clearly explain each problem you identified and how you solved it.**
2. **Add relevant screenshots as proof of your work.**
3. **If possible, include short screen-recording videos for major tasks.**
4. **Show the before and after status wherever applicable.**
5. **Briefly explain what you did in simple language, avoiding unnecessary technical terms.**
6. **Organize the evidence according to each step of the practical task.**
7. **Make sure every solution is properly verified and documented.**
8. **Submit the final report as a PDF, along with the relevant screenshots and videos.**

The report should clearly demonstrate: Problem → Solution → Verification → Final Result.